বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি, কুমিল্লা

BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

EXAMINATION CONFIDENTIAL

Final Examination, SPRING 2026

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: CSE 223

Course Title: Digital Logic Design

Credit Hour: 03

Exam Duration: 3 Hours

Full Marks: 150

Notes:

a. Each question carries **30** marks.
b. **Answer all of the following FIVE questions.**
c. The figure on the right of each question indicates marks for the respective question.
d. Use a different answer script for each part.

The Course Learning Outcomes (CLOs) are:

CLO1: Explain the characteristics of Number system, Boolean algebra and different types Of combinational and sequential circuits.

CLO2: Apply Boolean, De Morgan's, and other related theorem to solve digital logic circuits.

CLO3: Design various combinational and sequential circuits.

Part-A

1. a. Compare between analog and digital systems by their signal representation, performance, noise immunity, accuracy, and applications. [CLO1→C3] 10

   b. A communication system stores data in hexadecimal but transmits it in binary. Convert $(3AF)_{16}$ to binary and explain why hexadecimal notation is preferred for representing binary data. [CLO1→C4] 10

   c. From the numbers 15_{10} and 7_{10} perform binary subtraction using the 1's complement and 2's complement methods. Compare the two methods with respect to their procedures and treatment of end-around carry. [CLO1→C4] 10

2. a. Explain the operation of a BCD-to-Excess-3 code converter with a suitable example. Design the corresponding combinational logic circuit using basic logic gates. [CLO3→C3] 15

   b. Analyze the operation of a 4-bit binary adder-subtractor and design the corresponding combinational logic. Verify its functionality with suitable input conditions. [CLO3→C4] 15

OR

Illustrate the limitations of a Ripple Carry Adder and explain how a Carry Lookahead Adder addresses these limitations. Construct the Carry Lookahead Adder circuit with a suitable logic diagram. [CLO3→C3]

3. a. Describe the complete design procedure of a combinational logic circuit from problem specification to hardware implementation with an appropriate example. [CLO1→C3] 10

b. Explain and verify how a Full Adder can be constructed using two half adders and an OR gate. [CLO1→C3] 10

c. Why is a Multiplexer called a **data selector**? Implement the following function with a multiplexer $F(A, B, C, D) = \pi(3, 8, 12)$ [CLO1→C4] 10

OR

What do you mean by **2048-bit** ROM? Construct a **32X4** ROM. [CLO1→C4]

Part-B

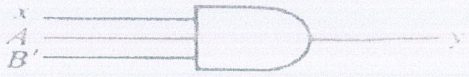
4. a. Compare SR, JK, D, and T flip-flops with respect to their characteristic tables, excitation tables, and applications. [CLO3→C3] 10

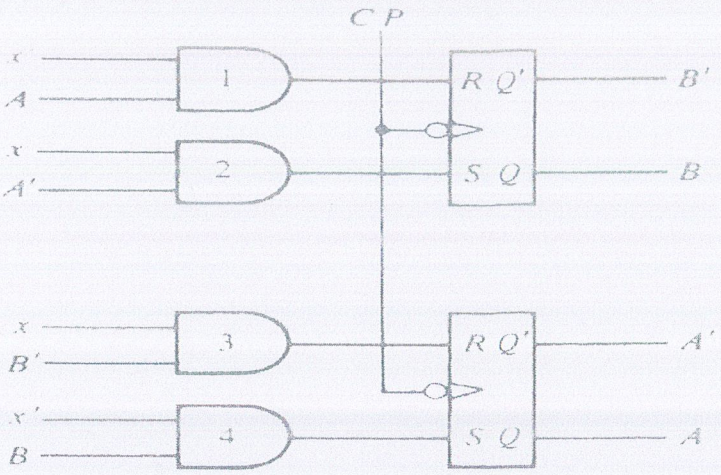
b. Explain the operation of a **Master-Slave SR Flip-Flop** with the help of a suitable block diagram and timing diagram. [CLO3→C3] 10

c. Describe the operation of a **D-type edge-triggered** flip-flop with **D=0, D=1** and **CP=0**. [CLO3→C3] 10

5. a. Given the following **clocked sequential circuit, derive:** [CLO3→C4] 20

- State Table
- State Diagram
- Next-State Equations





b. Write a short note on the following terms: [CLO3→C4] 10

i) Synchronous Counter

ii) Parallel Load Register

বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি (বাইউস্ট), কুমিল্লা
BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Term Final Examination, Spring 2026

Department of Computer Science and Engineering

Level-2 Term-II
Course Code: MATH -221
Course Title: MATH IV (Probability and Statistics)
Credit Hour: 03
Exam Duration: 03 hours
Full Marks -150

Notes:

a. Each question carries 30 marks.
b. Answer any five (05) questions. **Either three from Part-A and two from Part-B, or two from Part-A and three from Part-B.**

Part-A

1. a. Let X designate the number of heads obtained in the toss of three ideal coins. If two or more heads turn up, you receive Tk. 10 ; otherwise you pay TK. 5. 08

Find (a) The expected number of heads to turn up.

(b) The expected amount of money you would win. [CO2]

b. Consider the following joint probability distribution of X and Y: 08

$$f(x,y) = \frac{6}{5}\left(x+y^2\right), \quad for \quad 0<x<1, \quad 0<y<1$$
$$= 0, \quad otherwise$$

Calculate the following (i) $f(x/y)$, (ii) $f(x/0.5)$ and (iii) $P(Y<0.5/X=0.5)$.

[CO2]

c. The discrete random variable X and Y have the following joint probability distribution 14

	X		
Y	1	2	Total
0	1/8	1/8	2/8
1	2/8	4/8	6/8
Total	3/8	5/8	1

Estimate $E(X)$, $E(X/Y)$ and $E(Y/X)$. [CO2]

2. a. Twenty percent of the TVs produced in an industry are defective. If 4 TVs are placed in a box for marketing, **Find** that in how many boxes do you expect to have : **15**

i) one effective TV ;
ii) two defective TV ;
iii) at most 4 defective TVs in a consignment of 2000 such boxes ? **[CO2]**

b. Two discrete random variables X and Y have the following joint probability functions : **15**

	Values of X	
Values of Y	0	1
0	4/25	6/25
1	6/25	9/25

Compute $E(X)$, $E(Y)$, $E(X+Y)$ and $E(XY)$. Verify if X and Y are independent. **[CO2]**

3. a. The following data give the test scores and sales in lakh taka made by nine salesmen during the last year of a big departmental store. **15**

Test Scores	14	19	24	21	26	22	15	20	19
Sales	31	36	48	37	50	45	33	41	39

Find the regression equation on test scores on sales. **[CO1]**

b. The number of days of absenteeism of 80 workers of a factory over a particular year are recorded as follows: **15**

Days	0-3	4-7	8-11	12-15	16-19	20-23	24-27
Number of workers	5	14	17	25	10	6	3

If the authority decides to terminate 30% of the most irregular workers, **explain** how can you help the authority to take decision? **[CO1]**

Part-B

1. a. Two cricketers scored the following run randomly selected ten one-day matches. **15**

Player A	42	32	40	45	17	83	59	64	76	72
Player B	95	3	28	70	31	14	82	0	59	108

A prize is given to the best player. **Explain** who will get the prize? **[CO1]**

b. The following data 3, 4, 4, 5, 5, 7, 7 , 7, 12 relates to the marks obtained by nine students in a class. **Compute** mean deviation about mean, mean deviation about median, mean deviation about mode and comment. **[CO1]** 15

2. a. Suppose the average daily number of visitors to the BAIUST library has been 880 over the past several months. To investigate whether this average has changed recently, a random sample of 50 days is selected. The sample mean and standard deviation are found to be 870 and 21, respectively. If $\alpha = 0.05$, **compute** the power of the test and hence visualize the power curve. 18
N.B.: $Z_{\alpha/2} = 1.96, \phi(1.41) = 0.9297$ and $\phi(5.33) = 1$. **[CO2]**

b. The number of employees at six different drug stores of a small city are 4, 5, 6, 6, 7, 8. **Find** the variance and standard deviation of the data set taking it as population. 12

[CO2]

3. a. The following frequency distribution refers to the number of hours worked per month of 50 workers of a factory. **Estimate** Q_1, Q_3 and P_{65} and interpret the values you obtained. 15

Number of hours	30-55	55-80	80-105	105-130	130-155	155-180	180-205
Number of workers	3	4	6	9	12	11	5

[CO1]

b. The following frequency distribution refers to the number of hours worked per month by 50 workers of a factory. **Interpret** the variance and standard deviation of the frequency distribution by the short-cut method. **[CO1]** 15

Number of hours	30-55	55-80	80-105	105-130	130-155	155-180	180-205
Number of workers	3	4	6	9	12	11	5

t-test table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646

Percentage Points of the Chi-Square Distribution

Degrees of Freedom	Probability of a larger value of x^2								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28
5	0.554	1.145	1.610	2.675	4.351	6.63	9.24	11.07	15.09
6	0.872	1.635	2.204	3.455	5.348	7.84	10.64	12.59	16.81
7	1.239	2.167	2.833	4.255	6.346	9.04	12.02	14.07	18.48
8	1.647	2.733	3.490	5.071	7.344	10.22	13.36	15.51	20.09
9	2.088	3.325	4.168	5.899	8.343	11.39	14.68	16.92	21.67
10	2.558	3.940	4.865	6.737	9.342	12.55	15.99	18.31	23.21
11	3.053	4.575	5.578	7.584	10.341	13.70	17.28	19.68	24.72
12	3.571	5.226	6.304	8.438	11.340	14.85	18.55	21.03	26.22
13	4.107	5.892	7.042	9.299	12.340	15.98	19.81	22.36	27.69
14	4.660	6.571	7.790	10.165	13.339	17.12	21.06	23.68	29.14
15	5.229	7.261	8.547	11.037	14.339	18.25	22.31	25.00	30.58
16	5.812	7.962	9.312	11.912	15.338	19.37	23.54	26.30	32.00
17	6.408	8.672	10.085	12.792	16.338	20.49	24.77	27.59	33.41
18	7.015	9.390	10.865	13.675	17.338	21.60	25.99	28.87	34.80
19	7.633	10.117	11.651	14.562	18.338	22.72	27.20	30.14	36.19
20	8.260	10.851	12.443	15.452	19.337	23.83	28.41	31.41	37.57
22	9.542	12.338	14.041	17.240	21.337	26.04	30.81	33.92	40.29
24	10.856	13.848	15.659	19.037	23.337	28.24	33.20	36.42	42.98
26	12.198	15.379	17.292	20.843	25.336	30.43	35.56	38.89	45.64
28	13.565	16.928	18.939	22.657	27.336	32.62	37.92	41.34	48.28
30	14.953	18.493	20.599	24.478	29.336	34.80	40.26	43.77	50.89
40	22.164	26.509	29.051	33.660	39.335	45.62	51.80	55.76	63.69
50	27.707	34.764	37.689	42.942	49.335	56.33	63.17	67.50	76.15
60	37.485	43.188	46.459	52.294	59.335	66.98	74.40	79.08	88.38

বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি, কুমিল্লা

BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Term Final Examination, Spring 2026

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: CSE-225

Course Title: Numerical Methods

Exam Duration: 3 Hours

Full Marks: 150

Notes:

a. Figure on the right of each question indicate marks for respective question.
b. Answer **All (FIVE)** questions.
c. Use different answer scripts for each section.
d. Course Outcomes are

CO1: Recall the basics of accuracy and finite precision, round-off errors, truncation errors, and stability of numerical algorithms.
CO2: Apply appropriate numerical techniques and calculate the computational cost in the fields of science and engineering.
CO3: Formulate numerical techniques for root finding, curve fitting, differentiation, and integration.

Part-A

1. a. Determine the approximate value of $f(2.1)$ using Newton's Forward Difference Interpolation formula for $f(x) = 2x^3 - 4x + 1$, when $x_1 = 2$, $x_2 = 4$, and step value, $h = 0.25$. **[CO2→C3]** 15

b. Apply an appropriate interpolation method to determine the approximate number of students whose marks lie between 70 and 75 using the data provided in the following table. **[CO2→C3]** 15

Marks	30-40	40-50	50-60	60-70	70-80
No. of Students	31	42	51	35	31

OR

Construct the Forward Difference table from the given equally spaced data and apply appropriate interpolation method to estimate $f(2.5)$. Show all intermediate calculations. **[CO2→C3]**

x	2	3	4	5
$f(x)$	14.5	16.3	17.5	18

2. a. Solve the following system of linear equations using the Gauss Elimination Method. Show all intermediate steps and determine the values of the unknown variables. **[CO2→C3]** 15

$$2x - y + 3z = 9$$
$$x + y + z = 6$$
$$x - y + z = 2$$

b. Using the Gauss-Jordan method, obtain the approximate solution of the following linear system of equations: **[CO2→C3]** 15

$$x + y + z = 9$$
$$2x + 5y + 7z = 52$$
$$2x + y - z = 0$$

OR

Analyze the solution of the following system of linear equations obtained using the Matrix Inversion Method. Verify the correctness of the solution by substituting it into the original equations and comment on whether the coefficient matrix is invertible. **[CO2→C4]**

$$2x + y + z = 4$$
$$3x + 2y + 3z = 8$$
$$x + 4y + 9z = 14$$

3 a. Demonstrate the implementation of the Gauss quadrature method using the transformed form of the integral? Use the Gauss quadrature formula, find the solution of $\int_0^1 x^2 e^{-x} dx$ for $n = 2$. **[CO2→C3]** 20

b. Applying Simpson's 1/3 rule to estimate the integral: $\int_1^2 e^{x^2} dx$, take 4 intervals. **[CO2→C3]** 10

Part-B

4. a. Determine whether the Jacobi iterative method converges for the following system of linear equations by analyzing the convergence criteria. **[CO3→C4]** 20

$$27x + 6y - z = 85$$
$$6x + 15y + 2z = 72$$
$$x + y + 54z = 11$$

b. Apply the Gauss–Seidel iteration method to solve the given linear system, performing three iterations. **[CO3→C3]** 10

$$4x - 5y = -6$$
$$2x - 2y = 1$$

5 a. Consider an ordinary differential equation $\frac{dy}{dx} = x^2 + y^2$; $y(1) = 1.2$. Using the fourth order Runge–Kutta (R-K) method, find the value of $y(1.05)$ when $h = 0.05$. **[CO3→C3]** 20

b. Find the Taylor series of $x^3 - 10x + 5$ at $x = 2$. **[CO3→C3]** 10

বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি, কুমিল্লা
BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Term Final Examination, Spring 2026
Department of Computer Science and Engineering
Level-2 Term-II
Course Code: CSE-221
Course Title: Computer Algorithms and Complexity
Credit Hour: 03
Exam Duration: 3 Hours
Full Marks: 150

Notes:

a. Figure on the right of each question indicate marks for respective question.
b. Answer ***all FIVE*** questions.
c. Use different answer scripts for each part.
d. Course Learning outcomes are:
CLO1: Analyze the asymptotic performance and write rigorous correctness proofs for algorithms.
CLO2: Estimate the performance of various algorithms and data structures.
CLO3: Apply important algorithmic design paradigms and methods of analysis for problem solving.
CLO4: Device efficient algorithms in common engineering design situations.

Part-A

1. a. You are given a knapsack with a weight capacity of 15 units. You have the following items, each with a weight and a value. Apply the fractional knapsack algorithm to find the maximum value you can carry in the knapsack? **[CO3→C3]** **10**

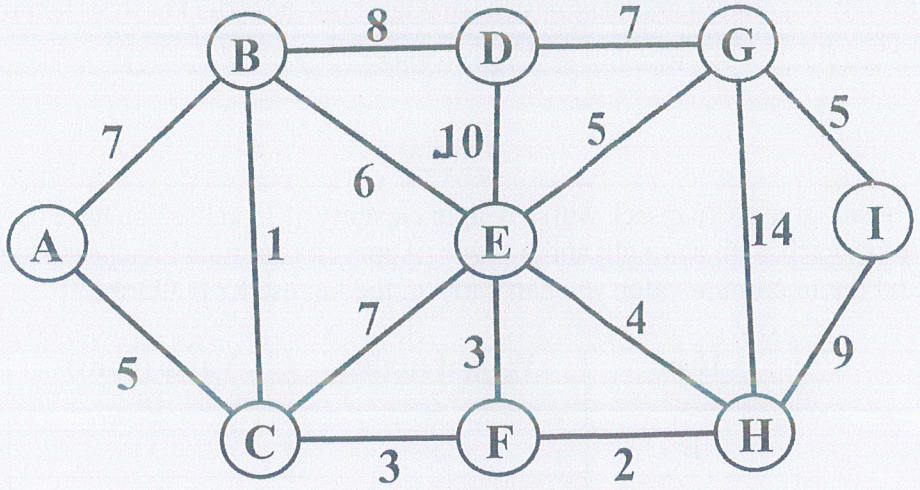
Item	Weight (units)	Value (units)
A	2	10
B	3	5
C	5	15
D	7	7
E	1	6
F	4	18
G	1	3

b. Consider the following message "**DABCCBBAADCCBBDAACD**" construct the Huffman tree and assign the corresponding binary codes to each character using Huffman Coding. **[CO3→C3]** **10**

c. You are given the following set of jobs, each with a deadline and a profit. Find the maximum profit using Job Sequencing with Deadline algorithm. **[CO3→C3]** **10**

Job	Profit	Deadline
J1	35	3
J2	30	4
J3	25	4
J4	20	2
J5	15	3
J6	12	1
J7	5	2

2. a. Apply Kruskal's and Prim's algorithms to the given connected, undirected, weighted graph to obtain the Minimum Spanning Tree (MST). Show the step-by-step edge/vertex selection process for each algorithm. **[CO3→C4]** **20**

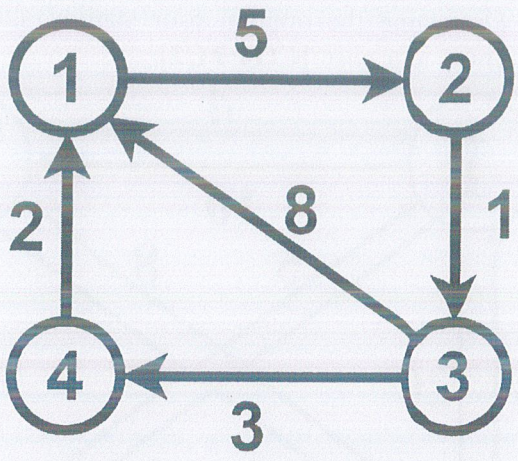


b. Given the connected, weighted graph with edge weights {(A, B, 4), (A, D, 3), (B, C, 2), (B, D, 5), (C, D, 1)}, apply Dijkstra's algorithm to determine the shortest path from node A to node D. Clearly show the intermediate steps, including the sequence of visited nodes, the updated distance table, and the final shortest path with its total cost. **[CO3→C3]** **10**

3. a. Develop the pseudocode for the Longest Common Subsequence (LCS) algorithm. Analyze the recurrence relation and the tracsseback process used to determine the LCS. **[CO4→C4]** **10**

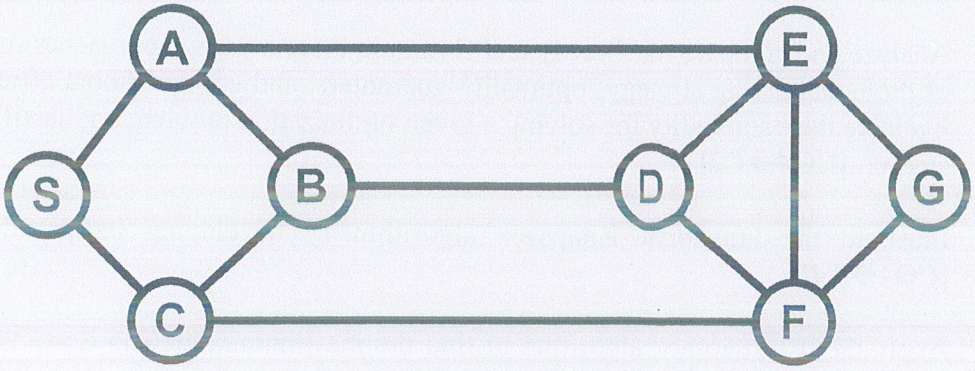
b. Given two sequences, X = {A, L, G, O, R, I, T, H, M} and Y = {L, O, G, A, R, I, T, H, M}. Find Longest Common Subsequence (LCS) using Dynamic Programming (DP). **[CO4→C3]** **10**

c. Implement the Floyd–Warshell algorithm to compute the shortest path between all pairs of vertices. Initialize the distance matrix using the adjacency matrix of the graph. Update the distance matrix iteratively using the Floyd–Warshell update rule. **[CO4→C4]** **10**



4. a. Write the pseudocode for the Depth-First Search (DFS) algorithm. Explain the traversal process and determine its time complexity. **[CO3→C4]** **10**

b. Perform a Breadth First Search (BFS) and Depth First Search (DFS) starting from vertex A. List the order in which the vertices are visited during the both traversal process. **[CO3→C3]** **20**

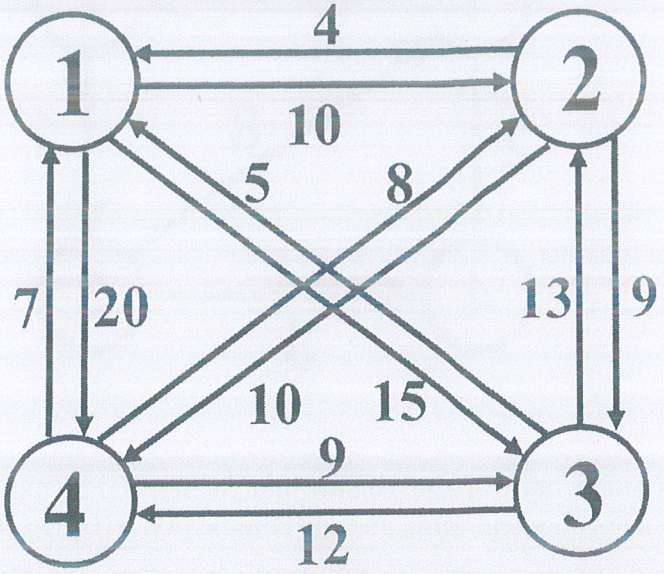


Part-B

5. a. Apply the Backtracking algorithm to solve the 4-Queens problem by finding all possible valid queen placements on a 4 × 4 chessboard where no two queens threaten each other. **[CO4→C3]** **10**

b. Demonstrate and justify why Dijkstra's algorithm fails in computing the shortest paths in graphs containing negative-weight edges. **[CO4→C3]** **10**

c. Using the Dynamic Programming approach, solve the Traveling Salesperson Problem (TSP). Determine the optimal tour starting and ending at city 1, and calculate the minimum tour cost. **[CO4→C3]** **10**



OR,

a. Analyze and compare the Greedy and Dynamic Programming approaches in terms of problem-solving strategy, optimality guarantees, and computational efficiency. Evaluate their suitability for solving a given optimization problem and justify your answer. **[CO4→C2]** **10**

b. Illustrate the Fractional Knapsack algorithm by developing its pseudocode. **[CO4→C4]** **10**

c. Apply the 0/1 knapsack algorithm to find the maximum value and items you can carry in the knapsack. with a bag of weight capacity of 8 units. You have the following items, each with a weight and a value: **[CO4→C3]** **10**

Item	Weight (units)	Value (units)
A	2	4
B	3	5
C	4	8
D	5	10

বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি, কুমিল্লা

BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Term Final Examination, Spring 2026

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: GED-221

Course Title: Engineering Economics

Credit Hour: 03

Exam Duration: 3 Hours

Full Marks: 150

Notes:

a. Figure on the right of each question indicates marks for the respective question.

b. Answer all of the following questions.

c. Use different answer scripts for each section

d. Course Learning Outcomes are:

CLO1: Develop preliminary knowledge of economic concepts for understanding of economy.

CLO2: Understand how microeconomic markets operate, identify the causes of market failure, and explain how government intervention influences market outcomes.

CLO3: Evaluate macroeconomic policies based on their effects on the economic variables of interest

CLO4: Apply different techniques of investment decisions.

PART-A

A1 i. Explain the concept of the consumption function and discuss its importance in macroeconomic analysis. [10]

ii. Analyze the macroeconomic determinants of consumption function. [10]

iii. A Savings function is given as, S= $-c + (1-d)\,Y$ [CLO3] [10]

a) Find out the consumption function.

b) Show that, APS+APC= 1

c) Show that, MPS+MPC=1

d) Find out the breakeven point.

A2 i. Define cost function. Draw a cost function graph showing the relationship among total cost (TC), total fixed cost (TFC), and total variable cost (TVC) and explain the relationships. [3+12]

ii. A cost function is given as $C= 100 + Q^3 - 10Q^2 + 40Q$. Calculate Total cost (TC), TVC, ATC, AVC and MC when a firm produces 10 units of output. [CLO2] [15]

EXAMINATION CONFIDENTIAL

A3 i. Define elasticity of demand. Demonstrate how cross elasticity values help distinguish between substitute and complementary goods using suitable examples. [3+12] [CLO1]

ii. When the price of tea increases from Tk. 50 to Tk. 60, the quantity demanded of sugar falls from 100 units to 130 units. Calculate the cross-price elasticity of demand and interpret the result. [15]

PART-B

B4. i. Define Game Theory. Discuss the concepts of Payoff of a Strategy, Dominant Strategy, Minimax Strategy, and Maximin Strategy, providing suitable examples for each. [2+8]

ii. Analyze the nature of interdependence in a duopoly market and explain the assumptions of Two-Person Zero-Sum Game. [10]

iii. [CLO2] [10]

		Firm II's Strategies				
		B_1	B_2	B_3	B_4	B_5
Firm I's Strategies	A_1	0.92	0.82	0.78	0.68	0.74
	A_2	0.58	0.62	0.48	0.42	0.54
	A_3	0.66	0.72	0.84	0.64	0.56
	A_4	0.76	0.88	0.70	0.46	0.82

Table: Payoff Matrix of Firm I

A Payoff Matrix of Firm I is given. Find out the Dominant strategies and Saddle point of Firm I and Firm II from payoff matrix of Firm I.

OR

i. Define Non-zero-sum game theory. Distinguish Zero-sum game from Non-zero-sum game. [15]

ii. "While the maximin strategy provides an optimal solution in the zero-sum game, this may not be so in the variable-sum game." Describe. [CLO2] [15]

B5. i. A production function is given as: $Q = 10L^{0.4}K^{0.6}$. Where, Q = Output, L=Labor and K= Capital. The firm employs **16 units of labor** and **25 units of capital.** [15]

a) Calculate the **marginal product of labor (MPL)** at this point. Interpret the economic meaning of the marginal product of labor in this context. **[CLO1]**

b) If the firm increases labor to 25 units (keeping capital fixed at 25), does the marginal product of labor increase or decrease? Justify using appropriate calculations or reasoning.

ii. Describe the three stages of Law of Variable Proportion. In which stage of production should a rational producer operate to maximize production? [6+9]

EXAMINATION CONFIDENTIAL